Maharashtra Education Society's
Institute of Management and Career Courses (IMCC), Pune
AUTONOMOUS

Academic Year 2026-27
FY MCA- SEM I
IT12 Data Structure and Algorithms
Practical Assignment1

Date: 15/10/2026 **Marks: 40**
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Create a Singly Linear Linked List with following operations

1. Create Linked List (append)
2. Traverse and print the node values
3. Insert node at a specific position
4. Delete alternate nodes in the linked list

CODE :

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

class SinglyLinkedList:
    def __init__(self):
        self.head = None

    def append(self, data):
        new_node = Node(data)
        if self.head == None:
            self.head = new_node
            return
        current = self.head
        while current.next != None:
```

```
                current = current.next
        current.next = new_node

    def traverse(self):
        if self.head == None:
            print("Linked List is empty")
            return
        current = self.head
        while current != None:
            print(current.data, end=" ")
            current = current.next

    def insert_at_position(self, data, position):
        if position < 1:
            print("Invalid position")
            return
        new_node = Node(data)
        if position == 1:
            new_node.next = self.head
            self.head = new_node
            return
        current = self.head
        count = 1
        while current != None and count < position - 1:
            current = current.next
            count += 1
        if current == None:
            print("Invalid position")
            return
        new_node.next = current.next
        current.next = new_node
```

```
                def delete_alternate_nodes(self):
        current = self.head
        while current != None and current.next != None:
            current.next = current.next.next
            current = current.next

ll = SinglyLinkedList()

n = int(input("Enter number of nodes : "))
for i in range(n):
    value = int(input(f"node {i + 1}: "))
    ll.append(value)

ll.traverse()

value = int(input("\nEnter value to insert : "))
position = int(input("Enter position : "))
ll.insert_at_position(value, position)

ll.traverse()

ll.delete_alternate_nodes()

print("\ndeleting alternate nodes :")
ll.traverse()
```

OUTPUT :

```
PS C:\Users\ADMIN\Downloads> py .\linkedlist.py
Enter number of nodes : 5
node 1: 10
node 2: 20
node 3: 30
node 4: 40
node 5: 60
10 20 30 40 60
Enter value to insert : 25
Enter position : 3
10 20 25 30 40 60
deleting alternate nodes :
10 25 40
PS C:\Users\ADMIN\Downloads>
```